

2.3 Assess Algorithm Efficiency

کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 05 of 16

★ Golden

2.3 Assess Algorithm Efficiency ★ Golden

LEARN IT

Algorithm efficiency is measured in terms of **time** and **space** — how fast it runs and how much memory it uses.

The key considerations for assessing efficiency are:

1. **Number of steps** (described with **Big O notation**).
2. **Number of conditions** (IF decisions).
3. **Number of repetitions** (loops).

Time (speed) + Space (memory) → Efficiency

WORKED EXAMPLE

Worked example: two ways to find a name in 1,000,000 sorted records.

Linear search checks about 1,000,000 names in the worst case (time grows with n). Binary search checks about 20 names (it halves the list each time). Both can be correct and clear. They are not equally efficient.

REMEMBER IT

Efficiency is always “as the input grows”. An algorithm that feels instant on 10 items can freeze on 10 million items if it is $O(n^2)$.

COMMON MISTAKE

Mistake: judging efficiency by how short the code looks.

Fix: a five-line nested loop can do n^2 work. Count steps, conditions and repetitions — not lines of text.

اردو میں سمجھیں

کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔ تین چیزیں دیکھی جاتی ہیں: قدموں کی تعداد (بگ او)، شرطوں کی تعداد، اور دہرائی (لوپ) کی تعداد۔ چھوٹے ان پٹ پر تیز لگنے والا الگورتھم بڑے ان پٹ پر سست پڑ سکتا ہے۔

CHECK YOURSELF

1. Efficiency is measured in terms of which two resources?
2. List the three key considerations named in the source.

Answers: 1. Time (speed) and space (memory). 2. Number of steps (Big O), number of conditions, number of repetitions.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Efficiency is mainly:

- a. Time and extra memory vs n
- b. How pretty the UI is
- c. How many comments
- d. Font choice

MCQ 2. As n grows, we care about:

- a. Growth class (Big O), not a 2 ms blip
- b. Only the constant 2 ms
- c. Only variable names
- d. Only colours

MCQ 3. Space efficiency is:

- a. Extra RAM/disk beyond the input
- b. Screen size
- c. PDF margins
- d. Cable thickness

MCQ 4. An algorithm can be:

- a. Correct but inefficient
- b. Efficient but never correct and still useful always
- c. A pie
- d. A sprint

MCQ 5. This chapter's framework later uses:

- a. Steps, conditions, repetitions
- b. Only UX
- c. Only ER
- d. Only Figma

MCQ 6. Why efficiency is ★:

- a. Board likes compare-two-algorithms questions
- b. It replaces correctness
- c. It replaces HCI
- d. It replaces Python

Extra short questions

1. Define time and space efficiency.
2. Why not measure only one run on your PC?
3. Example of wasting space.
4. Example of wasting time.

Extra long question

1. Compare linear vs binary search on $n=2$, $n=1\,000\,000$ for time and space. State preconditions. 10-mark table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Time: how running time grows with input. Space: extra memory.
2. Machines differ; we need growth vs n.
3. Copying a huge list on every recursive call.
4. Linear search 1 million times on a static sorted list instead of binary/index.

Long

1. Time $\sim n$ vs $\sim \log n$. Space $O(1)$ both (iterative). Binary needs sorted. For $n=1e6$, $\sim 1e6$ vs ~ 20 comparisons. Correctness not sacrificed.